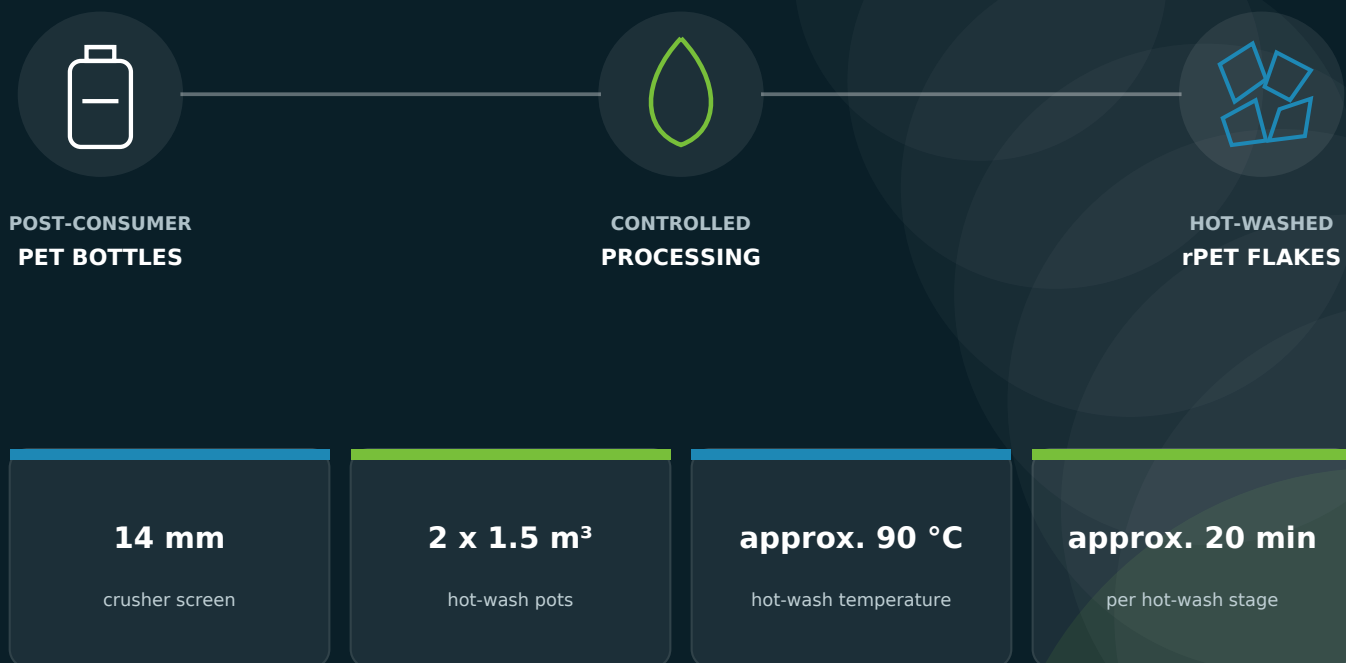


PROCESS OVERVIEW

From Post-Consumer PET Bottles to Hot-Washed rPET Flakes

A concise view of Poly Cleaner LLC's mechanical recycling line, process controls and batch traceability at its production facility in Baku, Azerbaijan.



01 MANUFACTURING FLOW

A documented route from incoming bottles to packed rPET flakes

The line combines manual material control, wet size reduction, density separation, two-stage hot washing, friction cleaning, rinsing, drying and final air separation.

01

Inbound and batch registration

Incoming post-consumer PET bottles are weighed, documented and assigned a batch identifier.

02

Pre-sorting and colour control

Large foreign materials and non-PET items are removed; bottles are separated by colour stream.

03

Label removal and metal control

A water-type label remover reduces label load before a second manual sorting and magnetic check.

04

Wet granulation

Bottles are size-reduced through a crusher fitted with a 14 mm screen; water supports initial cleaning.

05

Float-density separation

PET sinks while PP and PE caps, label residues and other light plastics float and are removed.

06

Two-stage hot washing

Two 1.5 m³ hot-wash pots operate at approximately 90 °C for about 20 minutes per stage, with high-speed friction washing between stages.

07

Rinsing dewatering and drying

Spray and rinse tanks remove residual chemicals and detached contamination before final dewatering and hot-air drying.

08

Final separation and packing

A zig-zag air separator removes light label and dust fractions; flakes move to pneumatic storage and big-bag filling.

FINISHED PRODUCT STREAMS

Clear / light-blue, green and mixed-colour hot-washed rPET flakes, released and supplied by batch according to grade and buyer specification.

02 CONTROL POINTS

Controls are applied from feedstock acceptance to final packing

Production and quality records connect the incoming batch, operating parameters, product checks and packed lot.

CONTROL POINT	WHAT IS CHECKED	REPRESENTATIVE RECORD
Inbound material	PET bottle stream, colour, foreign material, contamination and inbound weight	Acceptance and weight record
Sorting	PVC/PETG, metals, colour mix and other non-PET material	Shift record
Crusher output	Flake size and excessive fines	Process control form
Hot wash	Temperature, residence time and chemical dosing	Hot-wash log
Rinse output	pH and visual cleanliness	Quality control form
Finished flakes	Colour, labels/film, metal, PVC/PETG, moisture, bulk density and buyer-specific checks	COA / internal test protocol
Packing	Net weight, big-bag integrity, product grade and batch code	Warehouse record

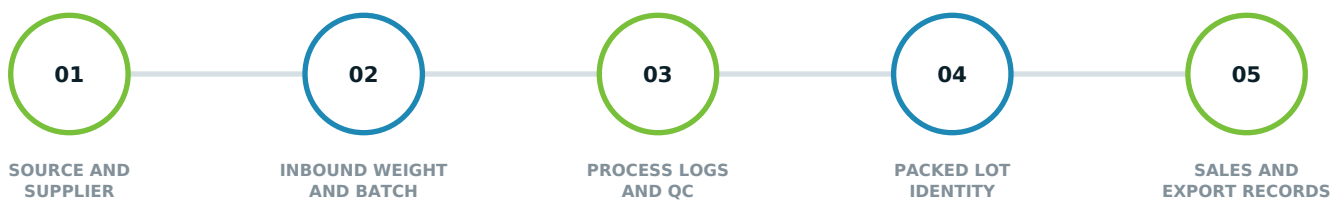
VERIFIED OPERATING PARAMETERS

SIZE REDUCTION 14 mm crusher screen	HOT WASH 2 x 1.5 m ³ pots	TEMPERATURE approximately 90 °C	RESIDENCE TIME approximately 20 min / stage
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03 BATCH ASSURANCE

Records follow the material through production and dispatch

Poly Cleaner's batch-based approach links feedstock receipt, process settings, quality review, packed product and commercial documentation.



FINISHED PRODUCT REVIEW

- **Colour stream**
Clear / light-blue, green or mixed-colour grade identification
- **Non-PET control**
PVC/PETG, metals and other foreign material checks
- **Surface cleanliness**
Label, film, dust and visible contamination review
- **Physical properties**
Moisture and bulk density checks
- **Buyer alignment**
Additional testing against the applicable customer specification
- **Release control**
Sampling and quality review before the lot is supplied

SEPARATED MATERIAL STREAMS

PP/PE caps, labels and films, metals, glass, mineral contamination and other removed fractions are collected separately and directed to an appropriate recovery or disposal route.

COMMERCIAL OUTPUT

CLEAR / LIGHT-BLUE

Hot-washed rPET flakes for applications agreed with the buyer.

GREEN

A separated colour stream for colour-controlled conversion.

MIXED COLOUR

A commercial stream for suitable non-clear industrial applications.

Product specifications and acceptance parameters are confirmed in the applicable quotation, technical data sheet and/or agreed buyer documentation. This overview does not make a food-contact compliance claim.